

OLLAMA COMMANDS AND THEIR FUNCTIONS

Introduction

Ollama is a platform that allows users to run Large Language Models (LLMs) locally on their computers. It provides simple commands to download, manage, and interact with AI models such as Llama 3, Mistral, Gemma, and DeepSeek.

Objectives

Understand the basic Ollama commands.
Learn how to download and run AI models.
Manage installed and running models.
Configure model parameters.

Commands and Descriptions

1. **ollama --version**

Purpose: Displays the installed Ollama version.
Command:ollama --version

2. **ollama help**

Purpose: Shows all available Ollama commands.
Command:ollama help

3. **ollama serve**

Purpose: Starts the Ollama server.
Command:ollama serve

4. **ollama pull**

Purpose: Downloads a model from the Ollama library.
Command:ollama pull llama3

5. **ollama list**

Purpose: Displays all installed models.
Command:ollama list

6. **ollama run**

Purpose: Runs a model and starts a chat session.
Command:ollama run llama3

7. **ollama ps**

Purpose: Shows currently running models.
Command:ollama ps

8. **ollama show**

Purpose: Displays information about a model.
Command:ollama show llama3

9. **ollama stop**

Purpose: Stops a running model.
Command:ollama stop llama3

10. ollama rm

Purpose: Removes a model from the system.

Command:ollama rm llama3

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\91831>ollama ls
NAME          ID          SIZE      MODIFIED
llama3.2:3b    a80c4f17acd5  2.0 GB    2 hours ago

C:\Users\91831>ollama run llama3.2:3b
>>> hi
How can I assist you today?

>>> /bye

C:\Users\91831>ollama ps
NAME          ID          SIZE      PROCESSOR  CONTEXT  UNTIL
llama3.2:3b    a80c4f17acd5  2.6 GB    100% CPU   4096     4 minutes from now

C:\Users\91831>ollama show llama3.2:3b
Model
  architecture  llama
  parameters    3.2B
  context length 131072
  embedding length 3072
  quantization  Q4_K_M

Capabilities
  completion
  tools

Parameters
  stop "<|start_header_id|>"
  stop "<|end_header_id|>"
  stop "<|eot_id|>"

License
LLAMA 3.2 COMMUNITY LICENSE AGREEMENT
Llama 3.2 Version Release Date: September 25, 2024
...
```

USING POSTMAN TO RUN API REQUESTS

Introduction

Postman is a popular API testing tool used by developers to send requests to servers and analyze responses. It provides a user-friendly interface for testing REST APIs without writing code.

Objectives

Understand the basics of API testing.

Learn how to send GET and POST requests using Postman.

Analyze API responses.

Test local AI APIs such as Ollama.

Software Required

Postman

Internet connection (for online APIs)

Ollama (optional, for local AI API testing)

Procedure

Step 1: Open Postman

Launch the Postman application on your computer.

Step 2: Create a New Request

Click New.

Select HTTP Request.

Enter a request name and save it.

Step 3: Send a GET Request

Select GET from the request type dropdown.

Enter the API URL.

Example:

`https://jsonplaceholder.typicode.com/posts/1`

Click Send.

Step 4: View the Response

Postman displays:

Status Code

Response Time

Response Body (JSON)

Example Response:

JSON

```
{
  "userId": 1,
  "id": 1,
  "title": "sample title",
  "body": "sample content"
}
```

Step 5: Send a POST Request

Change the method to POST.

Enter the API URL.
Open the Body tab.
Select raw and choose JSON.
Enter JSON data.

Example:

JSON

```
{  
  "title": "Hello",  
  "body": "Testing Postman"  
}
```

Click Send.

Step 6: Test Ollama API

Ensure Ollama is running.

API URL:

`http://localhost:11434/api/generate`

Method:

POST

Body:

JSON

```
{  
  "model": "llama3",  
  "prompt": "What is Artificial Intelligence?",  
  "stream": false  
}
```

Click Send to receive the AI-generated response.

Results

Successfully sent GET requests.

Successfully sent POST requests.

Received JSON responses from APIs.

Tested local AI models through Ollama API.

Advantages of Postman

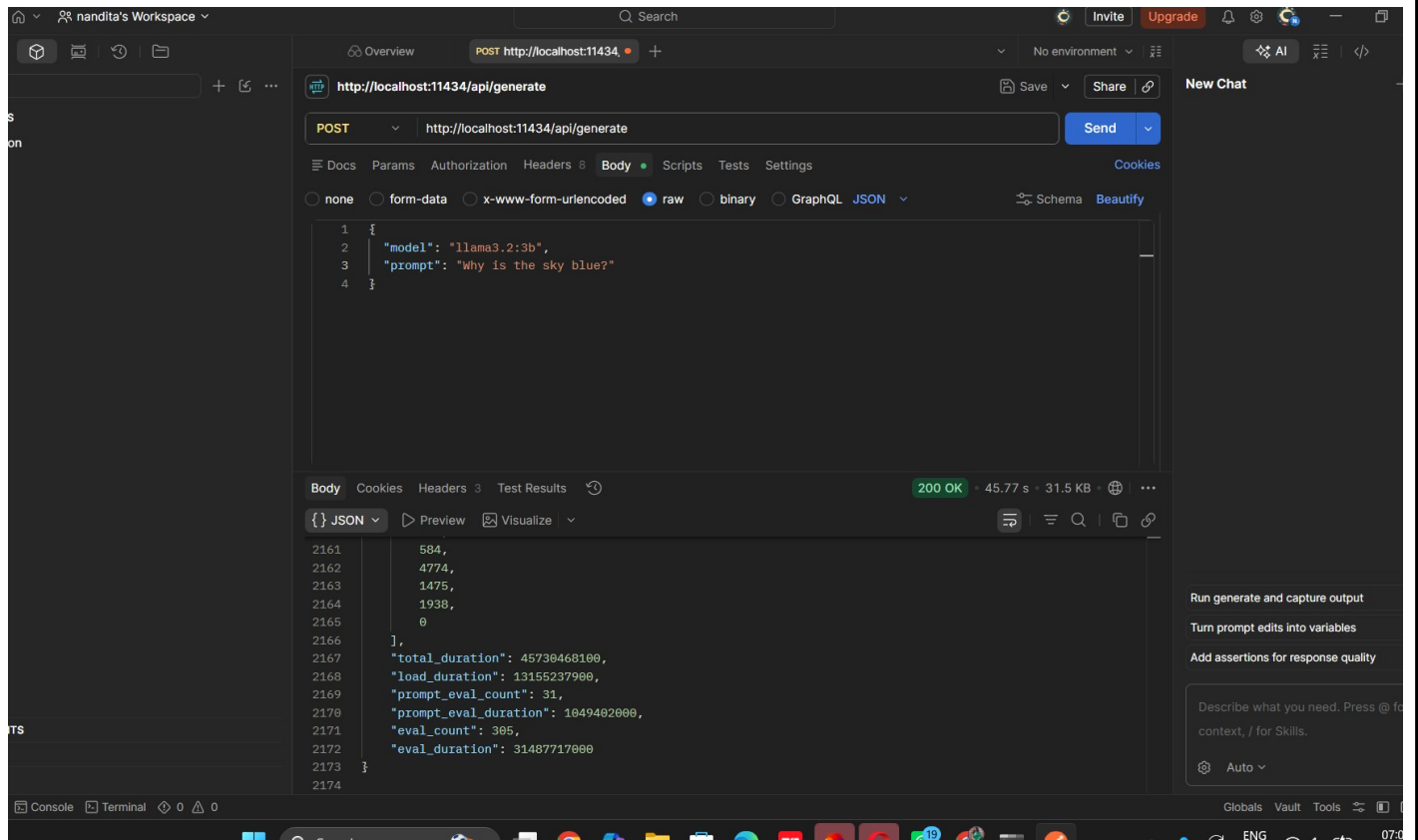
Easy-to-use graphical interface.

Supports all HTTP methods.

Allows API automation and testing.

Displays responses in a readable format.

Useful for debugging APIs.



Conclusion

Postman is an effective tool for testing and validating APIs. It simplifies sending GET and POST requests, analyzing responses, and testing local AI APIs such as Ollama without writing code. :::
This format is suitable for a lab report, internship diary, or practical record.

API Testing Using Python Requests and Ollama

Commands

1. Basic GET Request

Python

```
import requests
```

```
response = requests.get("https://jsonplaceholder.typicode.com/posts/1")  
print(response)
```

2. Check Status Code

Python

```
import requests
```

```
response = requests.get("https://jsonplaceholder.typicode.com/posts/1")  
print(response.status_code)
```

3. Display JSON Data

Python

```
import requests
```

```
response = requests.get("https://jsonplaceholder.typicode.com/posts/1")  
print(response.json())
```

4. Extract Title from JSON

Python

```
import requests
```

```
response = requests.get("https://jsonplaceholder.typicode.com/posts/1")  
data = response.json()
```

```
print(data["title"])
```

5. Random User API

Python

```
import requests
```

```
response = requests.get("https://randomuser.me/api/")  
print(response.json())
```

6. Dog API

Python

```
import requests
```

```
response = requests.get("https://dog.ceo/api/breeds/image/random")  
print(response.json())
```

7. Ollama POST Request

Python

```
import requests
```

```
url = "http://localhost:11434/api/generate"
```

```
data = {  
    "model": "llama3",  
    "prompt": "What is Python?",  
    "stream": False  
}
```

```
response = requests.post(url, json=data)
```

```
print(response.json())
```

```
IDLE Shell 3.13.11
File Edit Shell Debug Options Window Help
>>> = RESTART: C:\Users\91831\AppData\Local\Programs\Python\Python313\get_request.py
<Response [200]>
>>> = RESTART: C:\Users\91831\AppData\Local\Programs\Python\Python313\get_request.py
200
>>> = RESTART: C:\Users\91831\AppData\Local\Programs\Python\Python313\get_request.py
{'userId': 1, 'id': 1, 'title': 'sunt aut facere repellat provident occaecati excepturi optio reprehenderit', 'body': 'quia et suscipit\nsuscipit recusandae consequuntur expedit
m\nreprehenderit molestiae ut ut quas totam\nnostrum rerum est autem sunt rem eveniet architecto'}
>>> = RESTART: C:\Users\91831\AppData\Local\Programs\Python\Python313\get_request.py
sunt aut facere repellat provident occaecati excepturi optio reprehenderit
>>> = RESTART: C:\Users\91831\AppData\Local\Programs\Python\Python313\get_request.py
{'results': [{'gender': 'male', 'name': {'title': 'Mr', 'first': 'Mariano', 'last': 'Muñoz'}, 'location': {'street': {'number': 8963, 'name': 'Avenida de La Albufera'}, 'city': 'C
za', 'state': 'Comunidad de Madrid', 'country': 'Spain', 'postcode': 75026, 'coordinates': {'latitude': '-33.8081', 'longitude': '-143.7705'}, 'timezone': {'offset': '-6:00', 'c
ion': 'Central Time (US & Canada), Mexico City'}}], 'email': 'mariano.munoz@example.com', 'login': {'uuid': 'b93e2f7b-de5b-4bd3-a17f-88485cb5a046', 'username': 'heavyzebra568', 'c
d': 'alejandr', 'salt': 'Y4wTTYlh', 'md5': '9ab6f5068b0642689470bf7d55ed0c3e', 'sha1': '09671a707a57ae165a83eecd6f696979e8bcc670', 'sha256': 'bbc7aff516ed2425358c3b99775571aea22
9e58a23752d083f2aa5ad9'}, 'dob': {'date': '1973-01-11T22:17:15.902Z', 'age': 53}, 'registered': {'date': '2008-05-18T16:47:08.982Z', 'age': 18}, 'phone': '938-636-024', 'cell': '8-205', 'id': {'name': 'DNI', 'value': '33272959-N'}, 'picture': {'large': 'https://randomuser.me/api/portraits/men/54.jpg', 'medium': 'https://randomuser.me/api/portraits/med/m
pg', 'thumbnail': 'https://randomuser.me/api/portraits/thumb/men/54.jpg'}, 'nat': 'ES'}], 'info': {'seed': '99a819bbaacfab3e', 'results': 1, 'page': 1, 'version': '1.4'}}
>>> = RESTART: C:\Users\91831\AppData\Local\Programs\Python\Python313\get_request.py
{'message': 'https://images.dog.ceo/breeds/labrador/n02099712_619.jpg', 'status': 'success'}
>>> = RESTART: C:\Users\91831\AppData\Local\Programs\Python\Python313\get_request.py
{'error': 'model 'llama3' not found'}
>>> ===== RESTART: C:\Users\91831\AppData\Local\Programs\Python\Python313\get_request.py =====
{'model': 'llama3.2:3b ', 'created_at': '2026-06-11T14:21:46.3065133Z', 'response': 'Python is a high-level, interpreted programming language that is widely used for various pur
uch as web development, scientific computing, data analysis, artificial intelligence, and more. It was created in the late 1980s by Guido van Rossum and was first released in 19
Here are some key features of Python:\n\n1. **Easy to learn**: Python has a simple syntax and is relatively easy to read and write, making it a great language for beginners.\n2.
-level language**: Python is a high-level language, meaning it abstracts away many low-level details, allowing developers to focus on the logic of their code without worrying ab
ory management, etc.\n3. **Interpreted language**: Python code is interpreted line by line, which means that it does not need to be compiled before it can be executed.\n4. **Obj
ented**: Python supports object-oriented programming (OOP) concepts such as classes, objects, inheritance, and polymorphism.\n5. **Large standard library**: Python has a vast co
n of libraries and modules that make it easy to perform various tasks, such as data analysis, file I/O, networking, etc.\n\nSome popular use cases for Python include:\n\n1. **We
oment**: Python is used in web development frameworks such as Django and Flask.\n2. **Data science and analytics**: Python is widely used for data analysis, machine learning, a
visualization with libraries like NumPy, pandas, and Matplotlib.\n3. **Artificial intelligence and machine learning**: Python is used in AI and ML applications with libraries li
it-learn and TensorFlow.\n4. **Automation**: Python is often used to automate tasks such as data processing, file management, and system administration.\n5. **Scientific computi
python is widely used in scientific computing for tasks such as numerical analysis, signal processing, and visualization.\n\nOverall, Python is a versatile and powerful language
s become a popular choice for developers and researchers alike.', 'done': True, 'done_reason': 'stop', 'context': [128006, 9125, 128007, 271, 38766, 1303, 33025, 2696, 25, 6790,
366, 18, 271, 128009, 128006, 882, 128007, 271, 3923, 374, 13325, 30, 128009, 128006, 78191, 128007, 271, 31380, 374, 264, 1579, 11852, 11, 33398, 15840, 4221, 430, 374, 13882,
69, 5370, 10096, 1778, 439, 3566, 4500, 11, 12624, 25213, 11, 828, 6492, 11, 21075, 11478, 11, 323, 810, 13, 1102, 574, 3549, 304, 279, 3389, 220, 3753, 15, 82, 555, 12433, 78,
6870, 1264, 323, 574, 1176, 6004, 304, 220, 2550, 16, 382, 8586, 527, 1063, 1401, 4519, 315, 13325, 1473, 16, 13, 3146, 37830, 311, 4048, 96618, 13325, 706, 264, 4382, 20047, 3,
12309, 4228, 311, 1373, 323, 3350, 11, 3339, 433, 264, 2294, 4221, 369, 47950, 627, 17, 13, 3146, 12243, 11852, 4221, 96618, 13325, 374, 264, 1579, 11852, 4221, 11, 7438, 433, 6
, 3201, 1690, 3428, 11852, 3649, 11, 10923, 13707, 311, 5357, 389, 279, 12496, 315, 872, 2082, 2085, 40876, 922, 5044, 6373, 11, 5099, 627, 18, 13, 3146, 3386, 1762, 6702, 4221,
```

Using Ollama in Python

Step 1: Open Command Prompt

Check whether Ollama is installed:

```
ollama --version
```

Step 2: View Available Models

```
ollama list
```

Step 3: Download a Model (if not already installed)

```
ollama pull llama3
```

Step 4: Start Ollama

```
ollama serve
```

Keep this Command Prompt window running.

Step 5: Install Ollama Python Library

Open another Command Prompt:
pip install ollama

Step 6: Create a Python File
Create a file named:
ollama_test.py

Step 7: Import Ollama
Python
import ollama

Step 8: Generate a Response
Python
import ollama

```
response = ollama.generate(  
    model='llama3',  
    prompt='What is Python?'  
)
```

```
print(response['response'])
```

Step 9: Save the File
Save as:
ollama_test.py

Step 10: Run the Program
python ollama_test.py
Output:
Python is a high-level programming language...

Step 11: Chat Example
Python
import ollama

```
response = ollama.chat(  
    model='llama3',  
    messages=[  
        {  
            'role': 'user',  
            'content': 'Hello'  
        }  
    ]  
)
```

```
print(response['message']['content'])
```

Step 12: List Models in Python
Python
import ollama

```
print(ollama.list())
```


Python 3.13.11 (tags/v3.13.11:6278944, Dec 5 2025, 16:26:58) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>

= RESTART: C:\Users\91831\AppData\Local\Programs\Python\Python313\get_request.py

Python is a high-level, interpreted programming language that is widely used for various purposes such as web development, scientific computing, data analysis, artificial intelligence, and more. It was created in the late 1980s by Guido van Rossum and was first released in 1991.

Here are some key features of Python:

1. **Easy to learn**: Python has a simple syntax and is relatively easy to read and write, making it a great language for beginners.
2. **High-level language**: Python is a high-level language, meaning that it abstracts away many low-level details, allowing developers to focus on the logic of their program without worrying about memory management or other details.
3. **Interpreted language**: Python code is interpreted by an interpreter at runtime, rather than being compiled into machine code beforehand.
4. **Object-oriented**: Python supports object-oriented programming (OOP) concepts such as classes, objects, inheritance, and polymorphism.
5. **Large standard library**: Python has a vast collection of built-in modules and libraries that can be used for various tasks, including file I/O, networking, data structures, and more.
6. **Cross-platform**: Python can run on multiple operating systems, including Windows, macOS, and Linux.

Some common uses of Python include:

1. **Web development**: Python is often used with frameworks like Django and Flask to build web applications.
2. **Data analysis**: Python has libraries like NumPy, pandas, and matplotlib that make it easy to analyze and visualize data.
3. **Artificial intelligence and machine learning**: Python is widely used in AI and ML applications due to its simplicity and extensive libraries like scikit-learn and TensorFlow.
4. **Automation**: Python can be used to automate tasks, such as data scraping, file management, and more.
5. **Scientific computing**: Python has extensive support for scientific computing, including libraries like NumPy and SciPy.

Some of the key benefits of using Python include:

1. **Fast development**: Python's syntax is simple and easy to read, making it ideal for rapid prototyping and development.
2. **Large community**: Python has a massive and active community of developers who contribute to its ecosystem and provide support.
3. **Extensive libraries**: Python's standard library and third-party libraries make it easy to find solutions to common problems.

Overall, Python is a versatile and powerful language that is widely used in various industries and applications.

>>>